SOLUTION PROPOSAL DOCUMENT

DeceptIQ - Honey Cloud with AI Brain

Deception-Driven Threat Intelligence Platform — Team Cyfer Trace

1.0 Introduction

The Problem

Cyber-attacks on cloud infrastructure are growing faster than traditional detection systems can keep up. Conventional security tools like firewalls, Intrusion Detection System (IDS), Security Information and Event Management (SIEM) are reactive by nature. They wait for an attack to match a known signature before raising an alert.

This leaves organizations blind to novel attack techniques, zero-days and sophisticated threat actors who deliberately evade signature-based detection. Small and mid-size organizations are especially vulnerable because they lack the resources to run dedicated threat intelligence operations, leaving them exposed to threats they never even see coming.

Our Solution

Honey Cloud is a deception-driven threat intelligence platform that flips the defensive model on its head. Instead of waiting for attackers to breach real systems, Honey Cloud deploys a convincing fake cloud environment modelled as a cybersecurity firm managing blue team operations for multiple clients that actively lures, fingerprints, and studies attackers in real time.

When an attacker engages, a hybrid Machine Learning engine classifies their behaviour using both signature and behavioural detection while a GenAI-powered deception layer generates contextually convincing fake rewards to keep the attacker engaged long enough to build a complete threat profile.

Every interaction is enriched with live threat intelligence from AbuseIPDB and AlienVault Open Threat Exchange (OTX), visualized on a real-time dashboard, and exported as Structured Threat Information Expression (STIX) format intelligence turning every attack attempt into actionable security insight that benefits the entire threat intelligence community.

2.0 Existing Solutions and Their Limitations

Current approaches to honeypot and threat detection fall into three categories, each with critical gaps that Honey Cloud directly addresses.

Solution	What It Does	What It Misses
Traditional Honeypots	Logs attack attempts on fake services	Passive, no deception, no adaptation, easily fingerprinted
Signature-Based IDS/IPS	Detects known attack patterns	Blind to novel attacks, zero-days, and custom tooling
SIEM Platforms	Correlates logs and raises alerts	Reactive only, no attacker engagement or intelligence generation
Threat Intelligence Feeds	Provides known bad Internet Protocol and Indicator of Compromise data	No live attacker study, no behavioural profiling, static data
Commercial Honeypots	Deployed fake environments	Expensive, generic, no GenAI deception, not cloud-native

The common failure across all existing solutions is that they are built to react to attacks, not study attackers. None of them deceive, fingerprint, or generate fresh threat intelligence from live attacker behaviour. Honey Cloud fills this gap entirely.

3.0 Problem Statement

Blue Team Challenge

Create a sophisticated cloud-based honeypot system (Honey Cloud) to detect, analyze, and respond to cyber-attacks in real-time.

Key Objectives:

- Design a modular architecture for easy honeypot integration and scalability
- Use ML algorithms to analyze attack patterns and predict emerging threats
- Develop a comprehensive real-time reporting interface with threat visualization
- Explore collaboration mechanisms with cybersecurity firms and threat intelligence providers
- Implement adaptive honeypot environments that evolve based on attacker behaviour

Red Team Challenge

Launch sophisticated attacks against honeypot systems to test their detection and deception capabilities.

Key Objectives:

- Develop honeypot detection and fingerprinting techniques
- Create evasive attack strategies that avoid honeypot traps
- Test automated attack tools against adaptive honeypot systems
- Simulate advanced persistent threat (APT) scenarios
- Evaluate honeypot effectiveness across different attack vectors

4.0 Project Concept & Story

The Core Problem

- Traditional honeypots are passive, they log attacks and stop there
- Signature-based security tools miss novel and unknown attack techniques
- Small organizations lack resources for dedicated threat intelligence operations
- Attackers study their targets, defenders rarely study their attackers

The Idea

- Deploy a fake cloud environment that looks like a real, high-value target
- Lure attackers in, study their behaviour, deceive them, fingerprint them
- Turn every attack attempt into structured threat intelligence automatically

The Environment: Why a Fake Cybersecurity Firm?

- A security firm managing multiple clients is the highest-value target an attacker can find
- Breaching it means access to all their clients' infrastructure simultaneously
- This mirrors real-world supply chain attacks like SolarWinds
- Our team has genuine domain knowledge, the environment will be convincingly realistic

What Happens When an Attacker Arrives

- They find what looks like a real firm's cloud infrastructure, client portal, database, credentials
- System immediately fingerprints their tools, behaviour and skill level
- ML engine classifies the attack, known signature or novel behavioural pattern
- GenAI deception layer feeds them fake rewards to keep them engaged longer
- Threat intel feeds confirm if this attacker is a known global bad actor
- Full attacker profile is built, visualized, and exported as STIX threat intelligence

What Makes This Different

- Most honeypots watch attackers, Honey Cloud deceives them and learns from them
- The environment adapts based on attacker behaviour in real time
- ML and GenAI work together, a combination nobody else is building at this level
- Every attack makes the system smarter for the next one

5.0 Proposed Solution

Overview

Honey Cloud is a modular, cloud-based deception platform deployed on Google Cloud Platform (GCP) that attracts real attackers into a convincing fake environment, classifies their behaviour using a hybrid ML engine, deceives them using GenAI-generated responses, and exports structured threat intelligence automatically. The system is built in three tightly integrated layers: the Deception Layer, the Detection Layer, and the Intelligence Layer.

Layer 1: The Deception Layer (The Trap)

This is what the attacker sees and interacts with. The entire environment is designed to look like a real cybersecurity firm's cloud infrastructure.

- **Fake Web Application Backend:** A Flask-based client portal mimicking a real Blue Team operations dashboard with login pages, client listings, and security report previews
- **Exposed Database:** Honey DB instance loaded with fake but realistic client vulnerability assessments, SIEM credentials, and security incident reports
- **Leaked Cloud API Credentials:** Planted in a fake misconfigured GCP storage bucket, designed to be discoverable through standard attacker recon techniques
- **SSH Access Point:** Cowrie-based SSH honeypot skinned as the firm's internal server environment, not standalone but part of the larger convincing ecosystem

The key principle here is **context.** Individual fake services look suspicious in isolation. Together as one coherent firm environment, they look real.

Layer 2: The Detection Layer (The Brain)

Everything the attacker does inside the environment is captured, parsed, and analysed in real time.

Network & Endpoint Monitoring Pipeline:

- Zeek parses raw network traffic into structured connection logs
- Wazuh correlates endpoint-level events with network-level telemetry
- All events flow into PostgreSQL for correlation and forensic audit trail

Hybrid ML Engine (Two Models, One Output):

The two models run in parallel and their outputs are combined before reaching the dashboard.

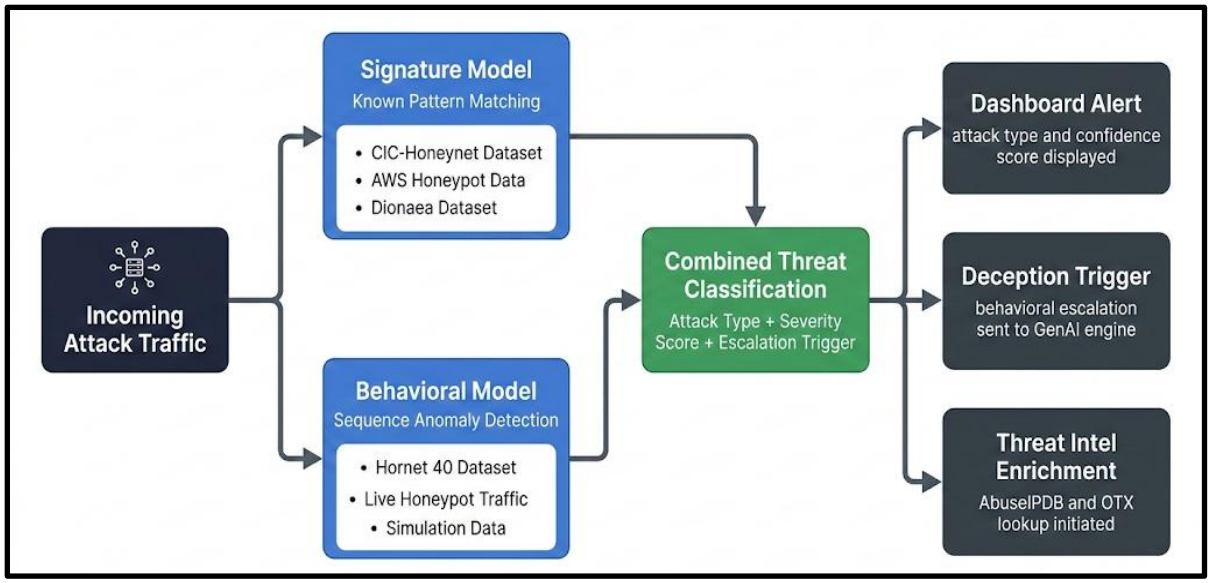


Fig 5.1: Hybrid ML Engine - Flow Chart

- **Signature Model:** Matches attack patterns against trained signatures from CIC-Honeynet, AWS Honeypot, and Dionaea datasets. Fast, reliable, catches common tools and known CVEs
- **Behavioural Model**: Analyses sequences of attacker actions over time, not individual events. An attacker who recons, then probes, then attempts credential stuffing is flagged differently than random noise. Trained on Hornet 40 dataset and live honeypot traffic.
- Both models feed a **combined threat classification output**, attack type, severity score, and a deception escalation trigger

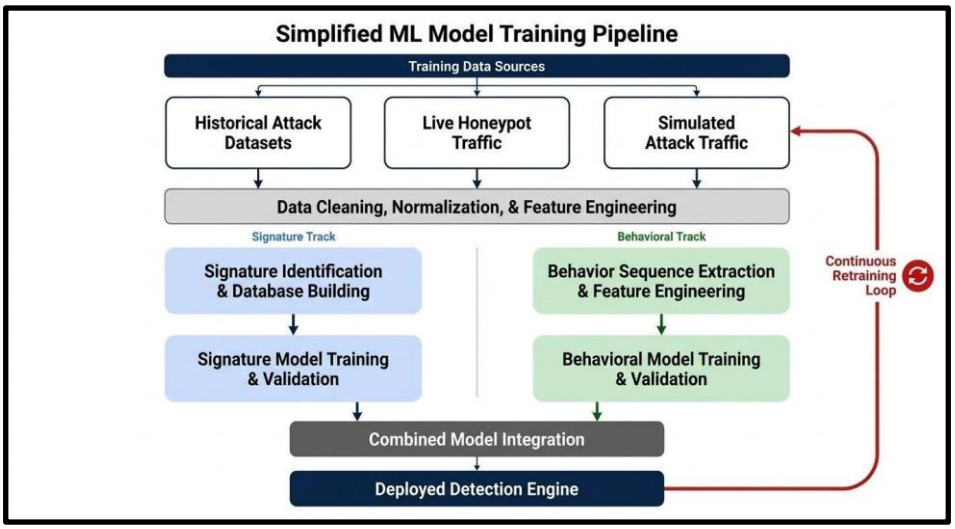


Fig 5.2: Hybrid ML Training - Flow Chart

Layer 3: The Intelligence Layer (The Response)

Once an attack is detected and classified, two things happen simultaneously: deception and intelligence enrichment.

Adaptive Deception Engine:
This is where ML and GenAI work together.

- The behavioural model decides **when** to escalate deception, detecting when an attacker is losing interest or about to leave
- Gemini Flash 2.0 via Vertex AI decides **what** fake content to generate, contextually convincing fake credentials, fake error messages, fake partial data access
- Ollama runs offline on GCP as a fallback, ensuring the deception engine never goes down during live operations or demo
- The result is **Progressive Deception**, attacker feels increasingly successful while the system builds an increasingly complete fingerprint

Threat Intelligence Enrichment:

Every attacker IP is automatically cross-referenced the moment a connection is established.

- **AbuseIPDB:** Primary check. Is this a known malicious IP? Confidence score returned instantly
- **AlienVault OTX:** Secondary enrichment. What campaigns, threat actors, and malware families are associated with this attacker?
- Combined enrichment feeds directly into the attacker profile on the dashboard

Attacker Fingerprint Profile includes:

- Tools and techniques used (Mapped with MITRE ATT&CK)
- Behavioural pattern and skill level assessment
- Known campaign associations from OTX
- AbuseIPDB reputation score
- Full attack timeline and sequence
- Deception interactions, what fake content was served and how attacker responded

Attack Simulation, Continuous Data Pipeline

A custom Python-based attack simulation script continuously launches known attack patterns against the honeypot. This ensures a steady stream of fresh training data for the machine learning models, even during periods when no real attackers are interacting with the system.

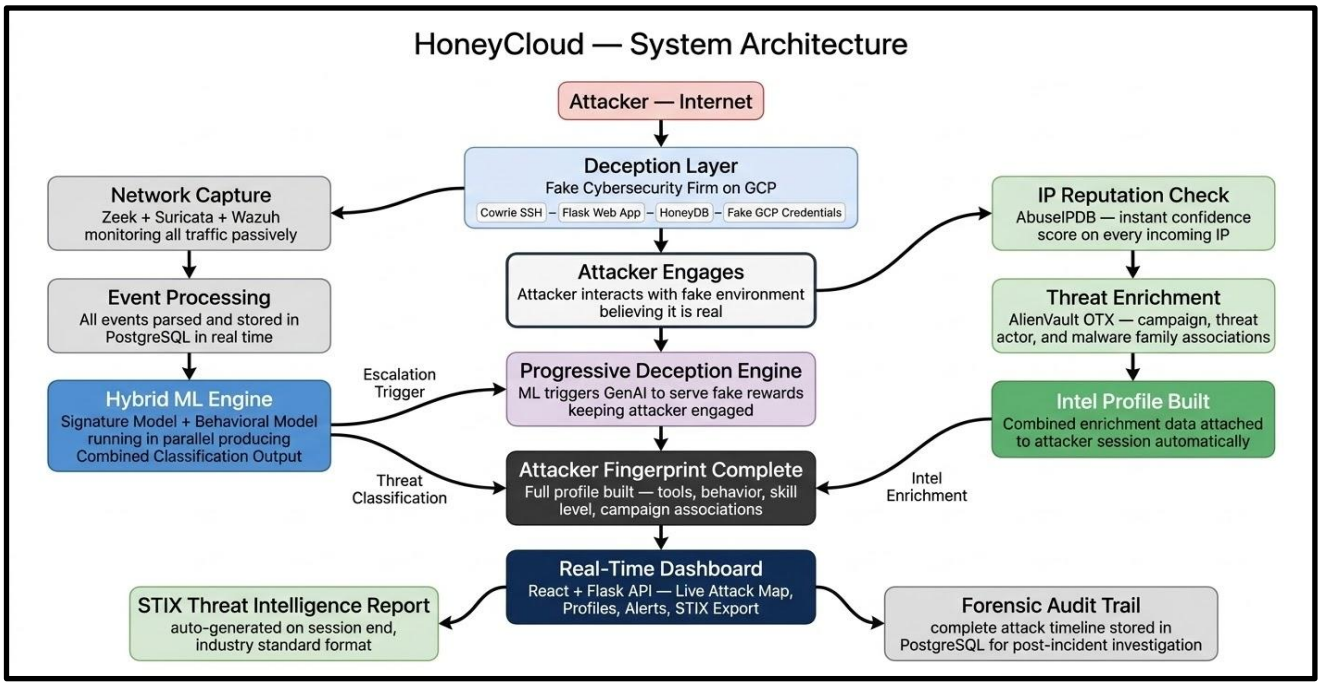
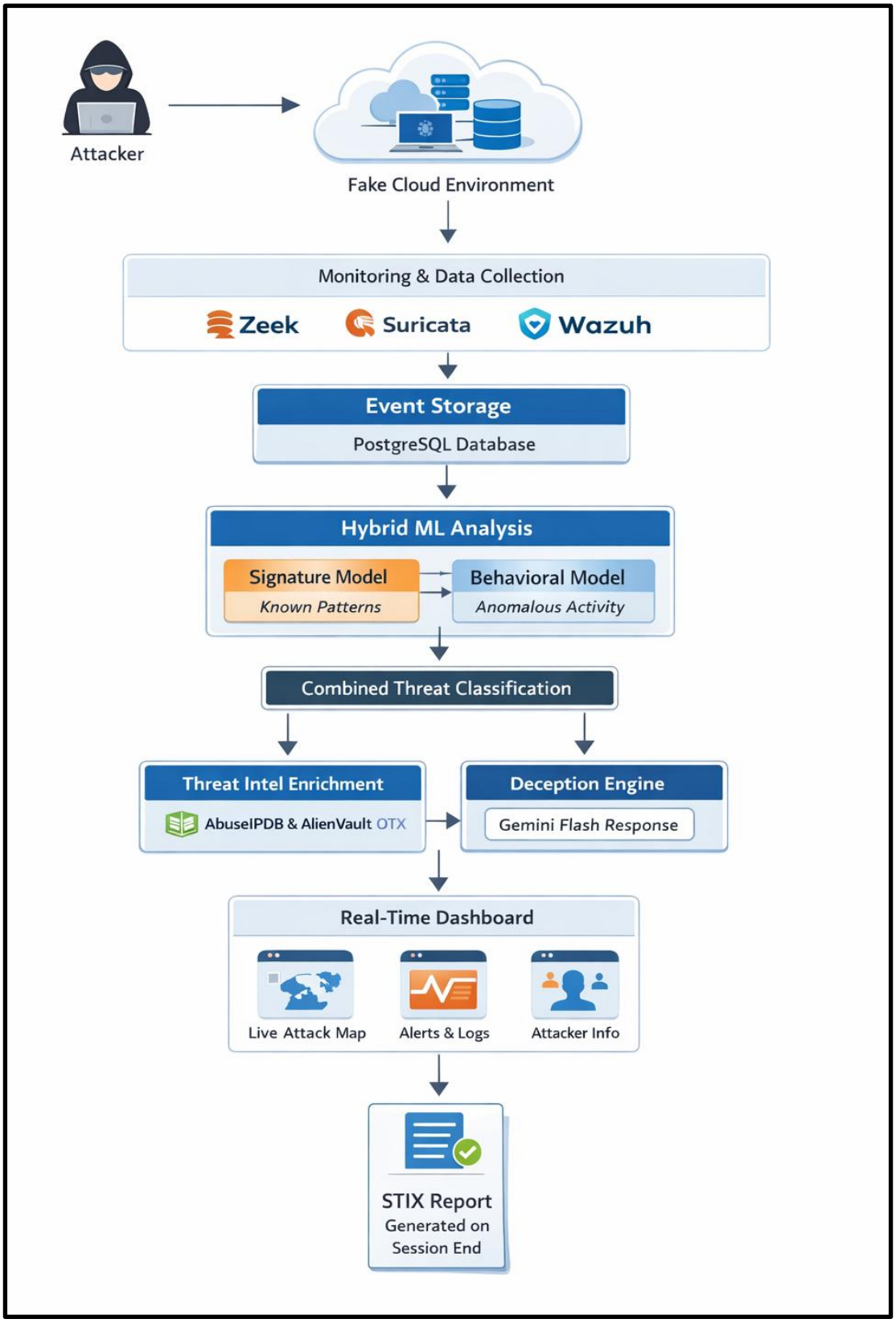


Fig 5.3: Honey Cloud System Architecture

Data Flow


Attacker
Fake Cloud Environment
Monitoring & Data Collection
Zeek
Suricata
Wazuh
Event Storage
PostgreSQL Database
Hybrid ML Analysis
Signature Model
Known Patterns
Behavioral Model
Anomalous Activity
Combined Threat Classification
Threat Intel Enrichment
AbuseIPDB & AlienVault OTX
Deception Engine
Gemini Flash Response
Real-Time Dashboard
Live Attack Map
Alerts & Logs
Attacker Info
STIX Report
Generated on
Session End

6.0 Four Core Features

Feature 1: The Honeypot Environment

A fake cybersecurity firm on GCP that looks real enough to fool actual attackers. One coherent environment, fake client portal, exposed database, leaked cloud credentials, and SSH access all telling the same believable story.

What keeps it convincing:

- **Progressive Deception:** Attacker gets fake partial wins to keep them going deeper into the environment
- **Camouflage Score:** Internally monitors how convincing the environment appears to each specific attacker and adjusts if suspicion is detected
- **Adaptive Content:** Fake responses, errors, and access levels adjust dynamically based on what the attacker is actively doing at that moment

Feature 2: Hybrid ML Detection Engine

Two models, two jobs, one combined output. Both run in parallel and their results are combined before reaching the dashboard.

- **Signature Model:** Catches known attacks, tools, and common vulnerabilities and exposures (CVEs) fast and reliably by matching patterns against trained signatures from multiple honeypot datasets
- **Behavioural Model:** Catches novel and unknown attacks by analyzing attacker action sequences over time, not individual events. An attacker who recons, then probes, then attempts credential stuffing is a fundamentally different signal than random noise

Combined output triggers dashboard alerts, deception escalation, and threat intel enrichment simultaneously the moment a threat is classified.

Feature 3: Adaptive Deception Engine

ML and GenAI working together in a closed feedback loop to deceive attackers dynamically in real time.

- **ML decides WHEN to escalate:** Detects when an attacker is losing interest, about to leave, or ready to go deeper and triggers the deception engine accordingly
- **Gemini Flash decides WHAT to generate:** Contextually convincing fake credentials, fake error messages, and fake partial data access that feels organic and specific to what that attacker is doing
- **Ollama runs offline on GCP:** Acts as fallback if Vertex AI experiences downtime, ensuring the deception engine never stops running during live operations or demo day

The result is an attacker who feels increasingly successful while the system simultaneously builds an increasingly complete fingerprint on them, automatically exported as a STIX threat intelligence report when the session ends.

Feature 4: Real-Time Threat Dashboard

The dashboard is the face of the entire project. Every feature built, detection, deception, threat intel, and attacker profiling, is experienced through this single interface. Built on React with a Flask API backend feeding live data continuously.

What It Shows:

- **Live Attack Map:** Geographic origin of every incoming attacker plotted in real time on a world map as connections are established
- **Active Threat Feed:** Live stream of attack events classified by type and severity as they happen
- **Attacker Profiles:** Per-attacker card showing tools used, behaviour pattern, skill level, AbuseIPDB reputation score and AlienVault OTX campaign associations
- **ML & Deception Panel:** Signature match or behavioural anomaly, attack classification, and confidence score per event, alongside what fake content was served, when, and how the attacker responded
- **Threat Intel Enrichment:** AbuseIPDB reputation and AlienVault OTX context displayed automatically for every attacker the moment they connect
- **STIX Export:** One click generates a structured, industry-standard threat intelligence report for any completed attacker session

7.0 Unique Features and Innovations

1. Fake Cybersecurity Firm as the Honeypot Environment - Most honeypots deploy generic fake services. Honey Cloud deploys a fully coherent fake cybersecurity firm managing Blue Team operations for multiple clients. This mirrors real-world supply chain attack scenarios, raises the perceived value of the target significantly, and attracts more sophisticated attackers than a generic SSH server or database ever would.

2. ML and GenAI Working Together for Live Deception - No other student-level project combines behavioural ML and GenAI specifically for honeypot deception. ML decides when to escalate, GenAI decides what to generate. The result is dynamic, contextually convincing fake responses that feel organic to the attacker rather than templated or scripted.

3. Progressive Deception with Camouflage Scoring - The system does not just serve fake content; it monitors how convincing the environment appears to each specific attacker in real time. If the camouflage score drops, the system adjusts. This is active deception management, not passive log collection.

4. Hybrid Two-Model Detection on Six Data Sources - Rather than one model trained on a single Kaggle dataset, Honey Cloud runs two specialized models in parallel trained across six data sources including live honeypot traffic and continuous red team simulation data. The combined output is significantly more accurate and harder to evade than single-model approaches.

5. Red Team Loop as a Core System Component - The automated attack simulation script is not just a demo tool. It continuously feeds fresh attack data into the ML training pipeline, validates detection rules, and ensures the system gets smarter every day without waiting for real attackers to arrive.

6. Production-Grade Reliability by Design - Ollama runs offline on GCP at all times as a fallback for Vertex AI. AbuseIPDB and AlienVault OTX run as parallel enrichment pipelines. STIX export meets the actual industry standard used by real threat intelligence platforms. Every architectural decision was made with production readiness in mind, not just demo performance.

8.0 Attack & Defence Techniques

Blue Team: Defence Techniques

Deception and Misdirection

- Deploying convincing fake environments to lure attackers away from real infrastructure
- Progressive deception feeding attackers fake partial wins to extend engagement and maximize fingerprinting time
- Adaptive content generation ensuring the environment stays convincing throughout the attack session

Detection and Monitoring

- Signature-based detection for fast, reliable identification of known attack tools and CVEs
- Behavioural sequence analysis catching novel and unknown attacks that signatures miss
- Network traffic analysis via Zeek and Suricata capturing every packet and connection in real time
- Endpoint correlation via Wazuh linking network events to system-level activity for complete forensic visibility

Threat Intelligence

- Real-time IP reputation checking via AbuseIPDB on every incoming connection
- Attacker enrichment via AlienVault OTX associating attackers with known campaigns and threat actors
- Automated STIX format threat intelligence export for industry-standard sharing and collaboration
- Attacker fingerprinting building complete behavioural profiles for future recognition and early warning

Incident Response

- Automated threat classification reducing analyst response time
- Complete attack timeline reconstruction from first connection to session end
- Forensic audit trail stored in PostgreSQL for post-incident investigation

Red Team: Attack Techniques

Reconnaissance

- Subdomain enumeration and endpoint discovery against exposed web services
- Port scanning and service fingerprinting to map the target environment
- Cloud storage bucket enumeration looking for misconfigured and publicly exposed credentials
- JavaScript analysis and API endpoint discovery on web application surfaces

Initial Access

- Credential stuffing against exposed login portals using leaked credential databases
- SSH brute force and dictionary attacks against exposed remote access services
- Exploiting misconfigured cloud API keys discovered during reconnaissance phase
- Web application attacks including SQLi, XSS, and CSRF against the fake client portal

Persistence and Lateral Movement

- Establishing persistent backdoor connections once initial access is achieved
- Using compromised credentials to move laterally across the fake environment
- Attempting privilege escalation through misconfigured service accounts and RBAC policies
- Simulating APT-style multi-stage attack chains across extended time periods

Evasion Techniques

- Honeypot fingerprinting and detection attempts to identify if the environment is a trap
- Low-and-slow attack patterns designed to avoid triggering behavioural anomaly thresholds
- Traffic obfuscation and tool customization to evade signature-based detection rules
- Rotating IP addresses and attack tooling to avoid reputation-based blocking

9.0 Application Area

Who This System Is Built For

Honey Cloud is designed for any organization that operates cloud infrastructure but lacks the resources, tools, or dedicated team to run active threat intelligence operations. The system operates entirely on GCP, integrates with existing security stacks, and requires no changes to production infrastructure since the honeypot environment runs completely separately as a deception layer.

Financial Systems and Fintech

- Users: Banks, payment gateways, UPI providers, fintech startups
- Environment: Cloud-hosted payment APIs, financial backends, customer data platforms

Healthcare and Medical Infrastructure

- Users: Hospitals, health record platforms, telemedicine providers
- Environment: Patient data systems, diagnostic platforms, medical record infrastructure

Web Applications and SaaS Platforms

- Users: SaaS companies, e-commerce, EdTech platforms
- Environment: Public-facing web apps, REST APIs, authentication systems

Cloud Infrastructure and DevOps

- Users: DevOps teams, cloud architects, managed cloud providers
- Environment: GCP, AWS, Azure environments with exposed storage, APIs, and pipelines

Cybersecurity Firms and MSSPs

- Users: SOC teams, threat intel analysts, Blue Team operations providers
- Environment: Multi-tenant security operations managing multiple client organizations

IoT and Critical Infrastructure

- Users: Smart city operators, energy sector teams, industrial automation engineers
- Environment: SCADA systems, industrial IoT networks, operational technology environments

10.0 Implementation Plan

Phase 1: Environment Setup and Data Collection

- Deploy GCP infrastructure and configure honeypot services (Cowrie, Glastopf, HoneyDB)
- Set up network monitoring pipeline (Zeek, Suricata, Wazuh)
- Acquire and preprocess training datasets (CIC-Honeynet, AWS Honeypot, Hornet 40, Dionaea)
- Tools: GCP, Python, PostgreSQL, Zeek, Suricata, Wazuh

Phase 2: ML Model Development

- Train and validate signature-based detection model on known attack datasets
- Train and validate behavioural anomaly detection model on sequence-based traffic data
- Integrate both models into a combined threat classification pipeline
- Tools: Python, Scikit-learn, PyTorch, Jupyter, PostgreSQL

Phase 3: Deception and Intelligence Integration

- Build adaptive deception engine connecting ML trigger to Gemini Flash response generation
- Integrate AbuseIPDB and AlienVault OTX threat intelligence APIs
- Implement attacker fingerprinting and STIX format export
- Tools: Vertex AI, Gemini Flash 2.0, Ollama, Python, Flask, AbuseIPDB API, AlienVault OTX API

Phase 4: Interface and Live Demo Preparation

- Build real-time React dashboard with live attack map, attacker profiles, and ML output panels
- Develop automated red team attack simulation script for continuous ML training and demo
- Full system integration testing and live demo rehearsal
- Tools: React, Flask, PostgreSQL, Python, GCP

11.0 Expected Outcome

Honey Cloud delivers a fully operational deception-driven threat intelligence platform that transforms passive defence into active attacker study. The final system achieves the following:

- Real-time detection and classification of both known and novel attack techniques against cloud infrastructure
- Accurate attacker fingerprinting including tools used, behavioural patterns, skill level, and threat actor associations
- Dynamic deception that keeps attackers engaged in a fake environment while building complete threat profiles automatically
- Continuous ML model improvement through live attack data and automated red team simulation
- Structured STIX threat intelligence reports generated automatically on session end, ready for industry sharing
- A live, demonstrable system that operates against real internet traffic on GCP with zero impact on production infrastructure

12.0 Tech Stack

Layer	Technology
Cloud Platform	GCP, and On-Premise Servers
Honeypot - SSH	Cowrie, Glastopf and Custom Flask, HoneyDB
Network Monitoring	Zeek and Suricata
Endpoint Monitoring	Wazuh
ML Models	Python, Scikit-learn, PyTorch
GenAI - Primary	Gemini Flash 2.0 via Vertex AI
Threat Intel	AbuseIPDB API, AlienVault OTX API
Backend & Database	Python and Flask, PostgreSQL
Frontend Dashboard	React
Threat Intel Export	STIX 2.1 Format

13.0 References

Datasets

1. **CIC-Honeynet Dataset:** Comprehensive honeypot data using T-Pot platform with 20+ honeypot types. https://www.honeynetproject.com/dataset.html
2. **Hornet 40 Dataset:** 40 days of geographically distributed honeypot traffic from 8 global locations. https://data.mendeley.com/datasets/tcfzkbpw46/3
3. **AWS Honeypot Attack Data:** 451,581 attack data points from cloud-based honeypots. https://www.kaggle.com/datasets/casimian2000/aws-honeypot-attack-data
4. **Cybersecurity Honeypot Attacks Dataset:** Dionaea honeypot data from AWS infrastructure: https://www.kaggle.com/datasets/startingsecurity/cybersecurity-honeypot-attacks

Open-Source Tools:

5. **Cowrie SSH Honeypot:** https://github.com/cowrie/cowrie
6. **Glastopf Web Application Honeypot:** https://github.com/mushorg/glastopf
7. **HoneyDB:** https://honeydb.io
8. **Zeek Network Security Monitor:** https://zeek.org
9. **Suricata IDS/IPS:** https://suricata.io
10. **Wazuh SIEM and XDR:** https://wazuh.com
11. **T-Pot Honeypot Platform:** https://github.com/telekom-security/tpotce

Threat Intelligence APIs:

12. **AbuseIPDB:** IP Address Abuse Reports and Reputation Checking. https://www.abuseipdb.com
13. **AlienVault OTX:** Open Threat Exchange Intelligence Platform. https://otx.alienvault.com

AI and Cloud Platforms:

14. **Google Vertex AI:** https://cloud.google.com/vertex-ai & **Gemini Flash 2.0:** Google DeepMind. https://deepmind.google/technologies/gemini
15. **Ollama:** Run Large Language Models Locally. https://ollama.com

Threat Intelligence Standards:

16. **STIX 2.1:** Structured Threat Information Expression. OASIS Open Cybersecurity Standards. https://oasis-open.github.io/cti-documentation/stix/intro
17. **TAXII 2.1:** Trusted Automated Exchange of Intelligence Information. https://oasis-open.github.io/cti-documentation/taxii/intro

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19. **MITRE ATT&CK Framework:** Adversarial Tactics, Techniques and Common Knowledge. https://attack.mitre.org
20. **OWASP Top 10:** Open Web Application Security Project. https://owasp.org/www-project-top-ten
21. **OWASP API Security Top 10:** https://owasp.org/www-project-api-security
22. **NIST Cybersecurity Framework:** https://www.nist.gov/cyberframework
23. **SolarWinds Supply Chain Attack Analysis:** CISA Advisory. https://www.cisa.gov/news-events/cybersecurity-advisories/aa20-352a

15.0 Conclusion

Honey Cloud is not just a honeypot. It is a deception-driven threat intelligence platform that actively studies attackers, adapts to their behaviour and transforms every attack attempt into structured, actionable intelligence. By combining a convincing fake cybersecurity firm environment, a hybrid ML detection engine, and a GenAI-powered deception layer, Honey Cloud addresses the fundamental weakness of every existing defensive solution; they react to attacks while Honey Cloud learns from them. The system is production-ready, cloud-native, and built to get smarter with every attacker that walks through the door.